KINES - LOWER EXTREMITIES (FINALS)

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TOPIC 1: HIP AND PELVIS COMPLEX

- Heavy weight-bearing anatomical region connecting the axial skeleton to the lower extremities.

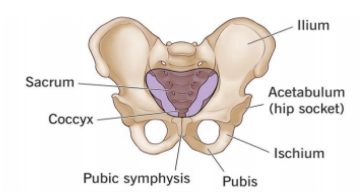
A. General Functions

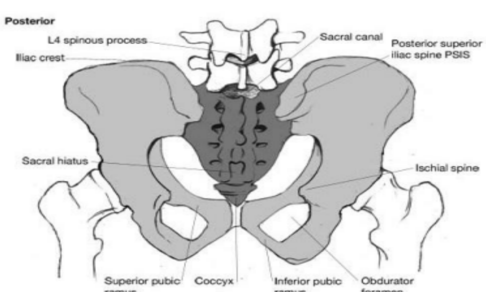
1. Weight bearing
2. Postural stability
3. Shock absorption
4. Force transmission (from trunk to lower limbs)
5. Mobility for gait and functional activities

B. Bones of the Pelvis & Pelvic Girdle

1. **Innominate Bones (Right and Left):** Formed by three fused bones:

ILIUM	Landmarks: - Iliac crest - ASIS, AIIS, PSIS, PIIS
ISCHIUM	Landmarks: - Ischial tuberosity (weight-bearing in sitting)
PUBIS	Landmarks: - Pubic symphysis - Superior pubic ramus, Inferior pubic ramus

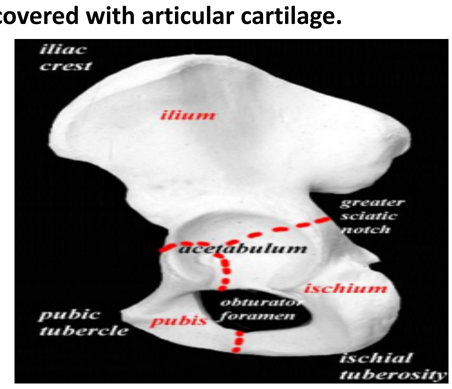




2. **Sacrum**
3. **Coccyx**

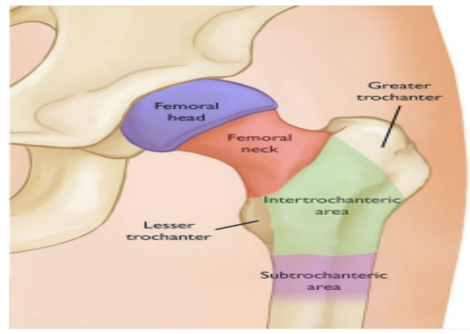
C. Acetabulum (Hip Socket)

- Formed by the **junction of the ilium, ischium, and pubis.**
- Serves as the **deep socket for the hip joint and is covered with articular cartilage.**



D. Proximal Femur Structures

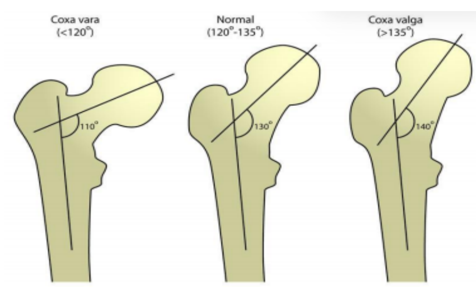
- Femoral Head
- Femoral Neck
- Greater Trochanter
- Lesser Trochanter
- Intertrochanteric Line (Anterior aspect)
- Intertrochanteric Crest (Posterior aspect)



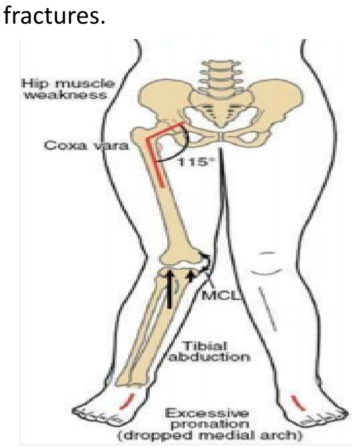
IMPORTANT ANGLES AND CLINICAL LINES OF THE HIP

Angle of Inclination (Neck-Shaft Angle)

- Formed between the **femoral neck and the femoral shaft in the frontal (coronal) plane.**



- **Normal Value - 125 degrees** in adults, **higher in children–150 degrees** (decreases with age)
- Variations:
  - **Coxa Vara -** decreased angle ***(<120 degrees)***
    - Femoral neck becomes more **horizontal**.
      - *Biomechanical Effects:* Increased bending stress on the femoral neck, shorter limb length, increased hip joint stability, and an increased load on abductors.
      - *Clinical Implications:* **Limping gait,** hip muscle weakness, tibial abduction, excessive pronation (**dropped medial arch**), and an increased risk of femoral neck fractures.



  - **Coxa Valga -** increased angle ***(>135 degrees)***

- Femoral neck becomes more vertical.
    - *Biomechanical Effects:* Reduced bending stress on the neck, longer limb length, decreased hip joint stability, and reduced abductor efficiency.
    - *Clinical Implications:* Hip instability and an increased risk of dislocation.

<u>Angle of Anteversion (Femoral Torsion Angle)</u>

- Angle measured in the **transverse (horizontal) plane** between the **axis of the femoral neck and the axis of the femoral condyles.**

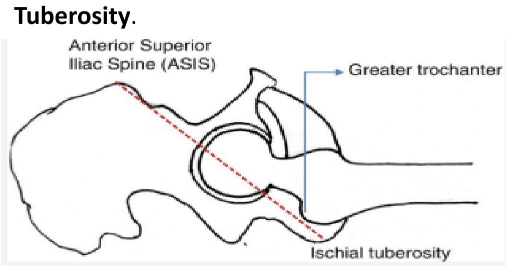
- **Normal Value - *10-15 degrees anterior***
- Variations:
    - **Increased Anteversion -** Femoral neck points more <u>anteriorly</u>.
        - *Effects:* Increased internal rotation of the hip and an ***"in-toeing"*** gait pattern.
        - *Common in:* Children (often physiologic) and individuals with cerebral palsy.
    - **Decreased Anteversion / Retroversion:** Femoral neck points more <u>posteriorly</u>.
        - *Effects:* Increased external rotation and an ***"out-toeing"*** gait pattern.

<u>Center-Edge Angle (CEA / Wiberg Angle)</u>

- Measures the **lateral coverage of the femoral head by the acetabulum**, reflecting lateral support and stability.
    - **Measurement:** Angle between a **vertical line through the center of the femoral head** and a **line extending from the femoral head center to the lateral acetabular edge.**
    - **Normal Value - *25-40 degrees***
        - **Low CEA:** Decreased stability with a high risk of dislocation and osteoarthritis.
        - **High CEA:** Excessive coverage with a high risk of femoroacetabular impingement.

<u>Nélaton's Line</u>

- An imaginary line drawn from the **ASIS** to the **Ischial Tuberosity**.



- **Normal Finding -** The greater trochanter of the femur lies exactly on or slightly below this line.
- **Clinical Use -** To assess the position of the greater trochanter in suspected hip pathology.
- **Abnormal Findings (Greater Trochanter ABOVE the line) -** Indicates the femur has shifted upward. Possible causes include femoral neck fracture, hip dislocation, coxa vara deformity, or shortening of the femur.
- **Abnormal Findings (Greater Trochanter BELOW the line) -** Rare; may suggest coxa valga or altered biomechanics.

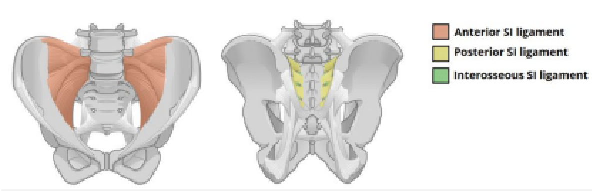
JOINTS AND LIGAMENTS OF THE PELVIS & HIP

<u>A. Hip Joint (Acetabulofemoral Joint)</u>

- **Synovial ball-and-socket joint** articulating the head of the femur with the acetabulum of the pelvis.
- **Labrum -** A **fibrocartilaginous rim** that **deepens the socket and enhances joint stability.**
- **Capsular Ligaments:**
    - **Iliofemoral Ligament (Y-Ligament of Bigelow) -** The <u>strongest ligament.</u>
        - Prevents hyperextension and provides robust anterior stability.
    - **Pubofemoral Ligament -** Limits **abduction and hip extension.**
    - **Ischiofemoral Ligament -** Limits **internal rotation.**
- ***Movements:*** *Flexion, extension, abduction, adduction, internal rotation, external rotation, and circumduction.*

<u>B. Sacroiliac Joint (SI Joint)</u>

- **Plane synovial joint** (with strong ligamentous support) with very limited mobility.
- **Function -** Shock absorption and force transmission between the upper body and lower limbs.
    - It **provides pelvic stability during walking and lifting.**
- **Stabilizing Ligaments -** Anterior sacroiliac, posterior sacroiliac, and the **interosseous sacroiliac ligaments <u>(the strongest)</u>.**
- **Movements (Minimal) -** Nutation (***anterior tilt of the sacrum***) and counternutation (***posterior tilt of the sacrum***).



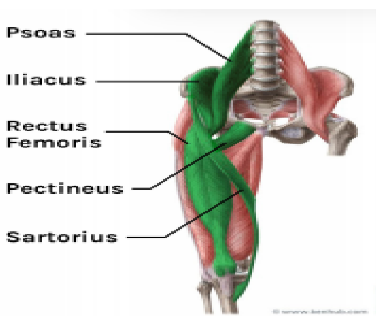
<u>C. Pubic Symphysis</u>

- **Secondary cartilaginous joint** (fibrocartilaginous joint) containing a **fibrocartilaginous disc** between the left and right pubic bones.
- **Reinforcing Ligaments -** Superior pubic ligament and inferior (arcuate) pubic ligament.
- **Function -** Provides stability to the pelvic ring and permits minimal movement for shock absorption.
- **Clinical Relevance -** Widens during pregnancy and childbirth; dysfunction results in anterior pelvic pain.

MUSCLE FUNCTION AND KINESIOLOGY OF THE HIP

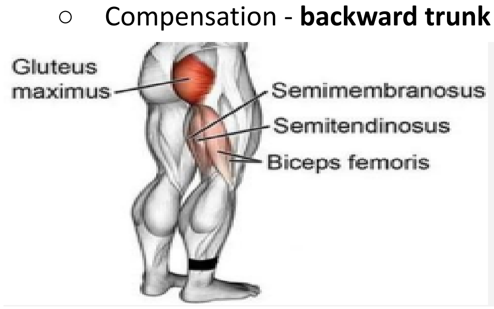
HIP FLEXORS - hip flexion, swing phase of gait, stair climbing

- **Iliopsoas** - primary hip flexor
- **Rectus femoris** - 2-joint muscle
- **Tensor Fascia Lata** - assists and stabilizes IT band
- Sartorius
- **Nerve supply:**
  - Femoral nerve (L2-L4)
  - Superior gluteal nerve (TFL)
- **Clinical**:
  - Tightness - **anterior pelvic tilt**
  - May lead to **low back pain**



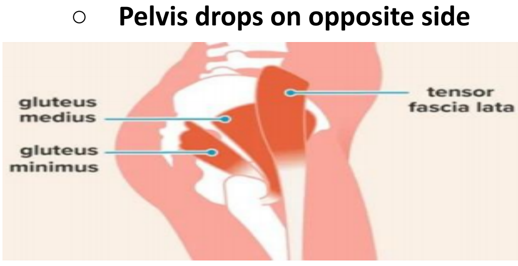
HIP EXTENSORS - hip extension, standing, climbing, and propulsion

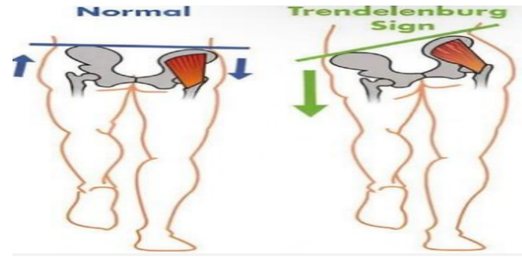
- **Gluteus Maximus** - most powerful extensor
- **Hamstrings** - assist in extension
- **Nerve supply:**
  - Inferior gluteal nerve (GMax)
  - Sciatic nerve (Hamstrings)
- **Clinical:**
  - Weakness - **difficulty standing**
  - Compensation - **backward trunk lean**



HIP ABDUCTORS - hip abduction, pelvic stabilization (single-leg stance), gait stability, prevents pelvic drop

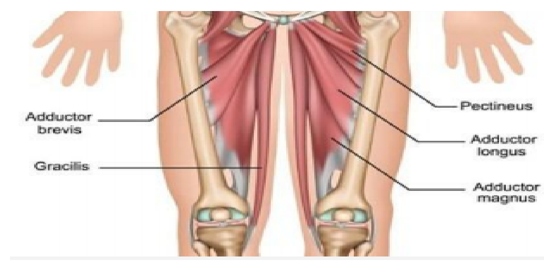
- Gluteus Medius, Gluteus Minimus, TFL
- **Nerve supply:**
  - Inferior gluteal nerve (GMax)
  - Sciatic nerve (Hamstrings)
- **Clinical**:
  - Weakness - **Trendelenburg gait**
  - **Pelvis drops on opposite side**





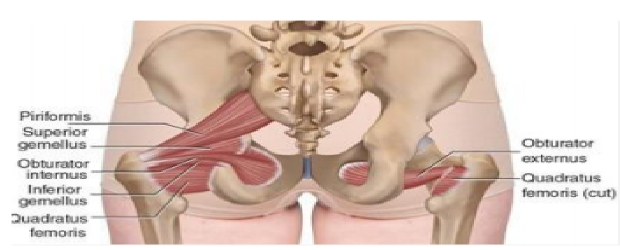
HIP ADDUCTORS - hip adduction, limb stabilization, control of medial movement

- **Adductor Longus, Adductor Brevis**
- **Adductor Magnus - strongest**
- **Gracilis - crosses hip and knee**
- **Pectineus**
- **Nerve supply:**
  - Obturator nerve (L2-L4)
  - Tibial part of sciatic nerve (adductor magnus - hamstring part)
  - Femoral nerve (pectineus, partial)
- **Clinical**:
  - Common injury - **groin strain**
  - Important for balance and gait control



HIP EXTERNAL ROTATORS - **external (lateral) rotation**, stabilization of femoral head; **deep stabilizers of the hip**

- **PGGOOQ** (Piriformis, Gemellus Superior and Inferior, Obturator Internus and Externus, Quadratus Femoris)
- Nerve supply:
  - Nerve to piriformis, nerve to obturator internus, nerve to quadratus femoris, obturator nerve (obturator externus)
- Clinical:
  - **Piriformis syndrome** - sciatic nerve compression
  - Important for joint stability during movement



HIP INTERNAL ROTATORS - **internal (medial) rotation**, assists in hip abduction, pelvic stabilization during gait

- **Gluteus Medius (anterior fibers), Gluteus Minimus** - primary internal rotators, GMed anterior fibers are for IR
- TFL - lateral stability via IT band
- **Nerve supply:**

- ○ Superior gluteal nerve

HIP LANDMARKS

Femoral Triangle

- **Boundaries -** Inguinal ligament (superior), Sartorius muscle (lateral), Adductor longus muscle (medial).
- **Floor -** Formed by the Iliopsoas, Pectineus, and Adductor longus muscles.
- **Contents (Mnemonic: NAVEL - from Lateral to Medial): N**erve (Femoral), **A**rtery (Femoral), **V**ein (Femoral), **E**mpty Space, **L**ymphatics.
- *Clinical Note:* **Cloquet's Node** is a specific lymph node located within the empty space if lymphatic congestion or pathology shifts into it.

Adductor Canal (Hunter's Canal / Subsartorial Canal)

- **Sheath Contents -** Artery, vein, empty space, and lymph nodes.
- **Canal Contents -** Empty space, lymph nodes.
- **Canal/Subsartorial Contents -** Saphenous nerve, femoral artery, femoral vein, empty space, and lymph nodes.
- **Ring Contents -** Naturally empty, unless occupied by an inguinal hernia.

BLOOD SUPPLY OF THE HIP

Main arterial sources:

1. **Medial Femoral Circumflex Artery (MFCA)**
    - Primary blood supply to femoral head and neck
    - Branch of deep femoral artery (profundus femoris)
    - Supplies the **posterior-superior portion of the femoral head**
- Clinical importance:
    - ○ Most critical artery
    - ○ Damage leads to high risk of ***AVN (Avascular Necrosis)***

2. **Lateral Femoral Circumflex Artery (LFCA)**
    - Also arises from deep femoral artery
    - **Supplies anterior portion of femoral neck and surrounding muscles**
    - **Less contribution** to femoral head

3. **Artery of the Ligamentum Teres (Foveal Artery)**
    - Branch of the obturator artery
    - Travels through ligament of head of femur
    - **Minor role in adults; important in children**

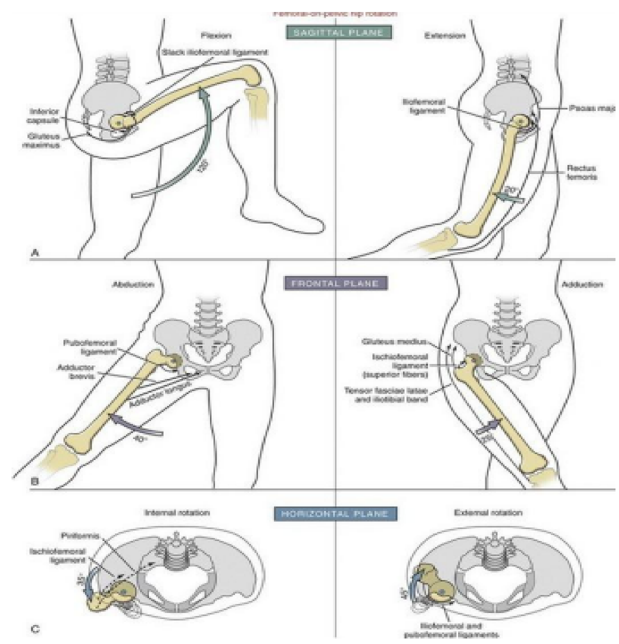
Retinacular arteries

- Branches of the MCFA
- Travel along femoral neck within the capsule
- **Directly supplies femoral head**
- Clinical importance:
    - ○ Easily damaged in fractures
    - ○ **Main source compromised in displaced femoral neck fractures**

HIP KINEMATICS AND RANGES OF MOTION (OSTEOKINEMATICS)

The hip joint possesses **three degrees of freedom**:

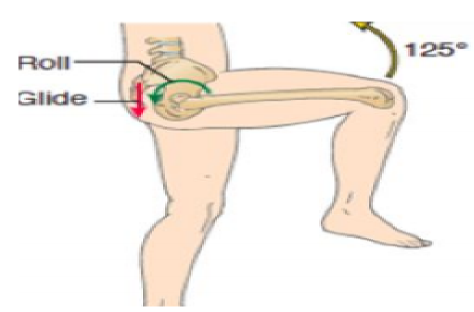
- **Sagittal Plane - Flexion *(0-120 degrees)*** with knee flexed; hitches slack iliofemoral ligament/inferior capsule/gluteus maximus)
    - **Extension *(0-20 degrees)*** tightens iliofemoral ligament/psoas major/rectus femoris.
- **Frontal Plane - Abduction *(0-45 degrees)*** tightens pubofemoral/adductor brevis/gluteus medius/superior ischiofemoral fibers)
    - **Adduction *(0-30 degrees)*** tightens adductor longus/TFL/IT band).
- **Transverse Plane - Internal Rotation *(0-35 degrees)***
    - **External Rotation *(0-45 degrees)***



Arthrokinematics of the Hip Joint

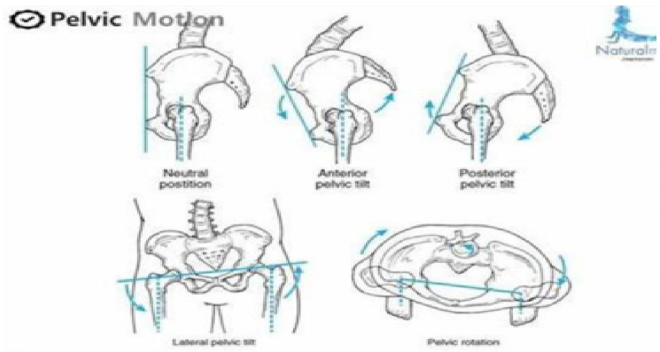
- Movement follows the rules of a **convex femoral head moving within a concave acetabulum**:

Flexion: Anterior roll, posterior glide. **Extension:** Posterior roll, anterior glide. **Abduction:** Superior roll, inferior glide. **Adduction:** Inferior roll, superior glide. **Internal Rotation:** Anterior roll, posterior glide. **External Rotation:** Posterior roll, anterior glide.



Pelvic Movements & Tilts

- **Anterior Pelvic Tilt -** ASIS moves anteriorly and inferiorly; **lumbar lordosis increases**.
    - ○ Associated with **sacral counternutation (sacral unlocking).**
- **Posterior Pelvic Tilt -** ASIS moves posteriorly and superiorly; **lumbar lordosis decreases.**
    - ○ Associated with **sacral nutation (sacral locking).**
- **Lateral Tilt -** One iliac crest **moves superiorly** relative to the other.
- **Pelvic Rotation -** Occurs dynamically during gait to improve step length and mechanical efficiency.

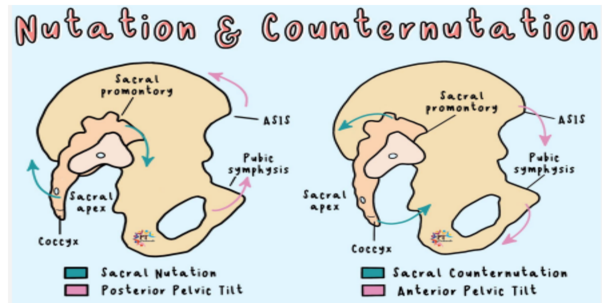


Open vs. Closed Kinematic Chain Mechanics of the Hip

Category	Closed Kinetic Chain (CKC)	Open Kinetic Chain (OKC)
Definition	Distal segment is **fixed**	Distal segment is **free to move**
Movement Pattern	Pelvis moves on a **fixed femur**	Femur moves on a **stable pelvis**
Joint Mechanics	Usually **weight-bearing**	**Non-weight-bearing**
Example Position	Standing, squatting	Sitting, supine leg raise
Hip Motion	Pelvic tilt, pelvic rotation	Hip flexion/extension, abduction/adduction
Muscle Activation	Co-contraction of multiple groups (stability)	Isolated muscle activation
Primary Function	Stability, balance, motor control	Mobility, isolated muscle strengthening
Force Transmission	High compressive forces across joints	Lower overall joint compression forces
Functional Relevance	Gait, sit-to-stand transitions, stair navigation	Kicking, lifting legs clear of obstacles

Pelvic-Sacral Force Mechanics: Nutation vs. Counternutation

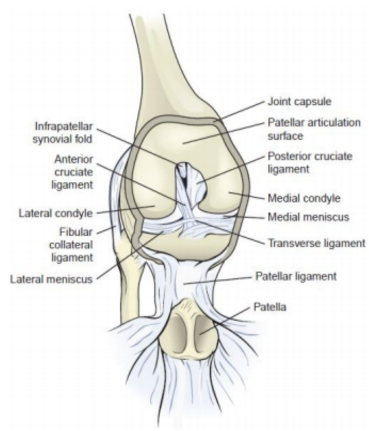
Feature	Nutation	Counternutation
Sacral Promontory	**Moves Anteriorly & Inferiorly**	**Moves Posteriorly & Superiorly**
Coccyx	**Moves Posteriorly**	**Moves Anteriorly**
Iliac Crest	Approximates (moves **closer together**)	Separates (**moves farther apart**)
Ischial Tuberosity	Separates (**widens pelvic outlet**)	**Approximates**
Pelvic Motion State	**ASIS is higher** than PSIS	**PSIS is higher** than ASIS
Pelvic Alignment Associated	Posterior Pelvic Tilt – **Sacral locking**	Anterior Pelvic Tilt – **Sacral unlocking**

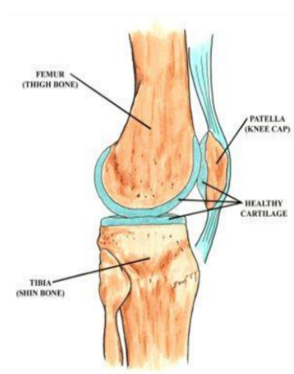


TOPIC 2: KNEE COMPLEX

KNEE - the **largest and among the most complex synovial joints in the body**

- **3 complex synovial joints with 3 bones:**
  - **Tibia**, **Femur**, and **Patella**.
    - Has **long lever arms** that render it structurally vulnerable, yet it can support the body in an erect position without active muscular effort.





Palpable Bony Features

- **Distal Femur -** Medial epicondyle, lateral epicondyle, medial condyle, lateral condyle, and the tibiofemoral joint line.
- **Proximal Tibia & Fibula -** Medial condyle or plateau, lateral condyle or plateau, tibial tuberosity, crest of the tibia, and the fibular head.
- **Patella -** Apex (inferior pole), base, anterior surface, and the posterior surface facets (lateral, medial, and the "odd" facet).
  - Boosts the **efficiency and torque** of the knee extension muscles.
  - Centers the **pull of the four quadriceps muscles** into a **single direction.**
  - Acts as a **smooth gliding mechanism** to lower friction and compression during deep bends.
  - Contributes to the **overall stability of the knee joint.**
  - Serves as a **bony shield** to protect the underlying thigh bone (femoral condyles) from direct impacts.

Note on Joint Definition: The proximal tibiofibular joint is **not** considered part of the knee joint complex.

JOINT ARTICULATIONS AND ARTHROKINEMATICS

1. Tibiofemoral Joint

- The medial and lateral femoral condyles are **distinctly convex both longitudinally and transversely.**
- They articulate with **two smaller tibial condyles that present only a very slight concavity.** Because of this structural incongruence, <u>knee flexion and extension cannot be pure rolling or hinge motions.</u>

- **Intercondylar Fossa -** Separates the **two femoral condyles inferiorly and posteriorly**, acting as the structural pathway through which the cruciate ligaments traverse.

- **Knee flexion - *125-150 degrees***
- **Knee hyperextension - *15 degrees***
- **Axial rotation** - The major functional importance of the motion is in closed chain motion in which **the femur rotates on the fixed tibia as in turning from kneeling, sitting, or squatting positions and in sudden changes in direction when running.**

- **Terminal Knee Extension**
  - When the knee moves into extension, the tibia laterally rotates about ***20 degrees*** on the fixed femur.
  - **Medial tibial rotation** occurs with **knee flexion and lateral tibial rotation with knee extension in OPEN CHAIN.**
  - **Medial femoral rotation** of the **femur on the fixed tibia, whereas lateral tibial rotation occurs during CLOSE CHAIN.**

DURING KNEE EXTENSION		DURING KNEE FLEXION	
OPEN CHAIN	CLOSED CHAIN	OPEN CHAIN	CLOSED CHAIN
TIBIA GLIDES ANTERIORLY ON FEMUR	FEMUR GLIDES POSTERIORLY ON TIBIA	TIBIA GLIDES POSTERIORLY ON FEMUR	FEMUR GLIDES ANTERIORLY ON TIBIA
From 20-30º knee flexion to full extension		From full knee extension to 20-30º flexion	
Tibia rotates externally	Femur rotates internally on stable tibia	Tibia rotates internally	Femur rotates externally on stable tibia

<u>Arthrokinematics of the Tibiofemoral Joint</u>

- **Close packed position** - full extension
- **Open packed position** - ***25 degrees*** of flexion

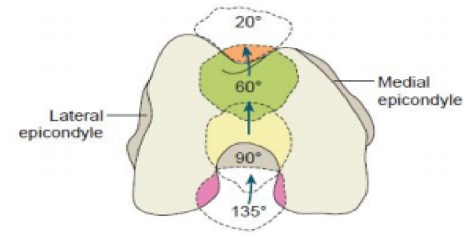
2. Patellofemoral Joint

- **Patella** - <u>largest sesamoid bone</u>
- **Quadriceps** - <u>actively stabilizes patella</u> on all sides and guides motion.
  - **VMO (Vastus Medialis Oblique)** - <u>dynamic stabilizer</u> (***0-20 degrees***)

<u>Kinematics of the Patellofemoral Joint</u>

- **Full Knee Extension -** The patella is **unrestricted and rests loosely at the proximal end of the intercondylar groove.**
  - It can be passively mobilized medially, laterally, superiorly, inferiorly, and rotated.
  - This serves as the **open-packed position**.

- **Knee Flexion progression -** As the knee flexes, the patella tracks inferiorly, progressively increasing the patellofemoral contact area until about ***90 degrees.***



- **Close-Packed Position - full knee flexion.**
- **Open-Packed position - full knee extension**

- **Q Angle -** An angle formed by **drawing a line from the ASIS to the center of the patella**, and a **second line extending from the tibial tuberosity upward through the patellar center.**
  - This angle is **<u>created by the natural adduction of the femoral shaft</u>**, which helps transmit body weight perpendicularly to the foot and ankle.
  - **Men - *10-14 degrees***
  - **Women - *15-23 degrees***

MUSCLE KINESIOLOGY & INSUFFICIENCIES OF THE KNEE

- **Knee Extensors -** for stabilizing, accelerating, and decelerating the knee.
  - The **Vastus Intermedius** acts as the <u>primary "workhorse".</u>
  - **VMO** - <u>medial patellar stabilizer</u>
  - Because the **Rectus Femoris** is a two-joint muscle, **it generates its greatest active extension torque when it is maximally elongated.**

- **Hamstrings -** Contract efficiently during prone trunk extension, the single-leg Romanian Deadlift (RDL), and when stabilizing the knee by **directly restricting an adverse anterior glide of the tibia on the femur.**
  - During the stance phase of gait, **they work with the sartorius and gracilis to control hip and knee rotation as the body moves forward.**

- **Popliteus -** Acts specifically to **unlock the knee** to initiate flexion from full extension.

- **Gastrocnemius -** A two-joint muscle that **dynamically performs simultaneous knee flexion and ankle plantarflexion.**

<u>Muscle Insufficiencies & Combined Movements</u>

1. **Knee Flexion + Hip Extension -** Dominated by the **hamstrings crossing two joints.**
   - This movement can be limited by **active insufficiency** of the hamstrings or flexibility limits of the anterior two-joint rectus femoris.

2. **Knee Extension + Hip Flexion - Active insufficiency of the rectus femoris** can prevent simultaneous full hip flexion and full active knee extension.
    - ***Gait Impact:*** If a straight leg raise is limited by muscle flexibility, it directly curtails step length and stride characteristics.

3. **Knee Flexion + Hip Flexion -** The hip flexors and hamstrings act synergistically to generate a highly efficient, useful motion utilized during walking, running, and hopping.

4. **Knee Extension + Hip Extension -** Used to propel the body upward or forward (e.g., rising from a chair, jumping, climbing stairs).
    - In a closed kinetic chain, this requires a powerful **co-contraction** of both the hamstrings and quadriceps.

5. **Knee Flexion + Ankle Plantarflexion -** Handled by the gastrocnemius performing both actions concurrently.

6. **Knee Extension + Ankle Plantarflexion -** Required functionally during actions like rising on tiptoes, running, and jumping.

7. **Extensor Lag -** The specific clinical inability to achieve full active knee extension despite possessing full passive knee extension range of motion.
    - It typically results from localized pain or severe quadriceps weakness.
    - *Note:* If full extension cannot be reached either actively or passively, it indicates a structural soft-tissue or joint restriction rather than a simple lag.

Pulley system of the Patella

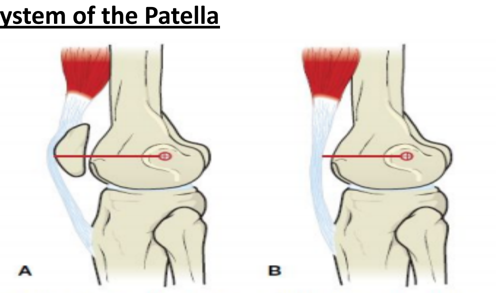


Figure 10.27 Pulley system of the patella and quadriceps.
A) With a patella the lever arm of the quadriceps is larger.
B) When the patella is absent, the moment arm of the quadriceps reduces significantly, causing a reduction in potential force provided by the quadriceps.

THE MENISCI

- **Fibrocartilaginous structures** that deepen the flat joint surface, improving overall structural congruency and stability.

- **Medial Meniscus -** C-shaped.
    - Anchored to the **lateral rim of the tibia** and **joint capsule by the coronary ligament.**
    - Its deep fibers **attach directly to the medial collateral ligament (MCL).**
    - The **semimembranosus muscle tendon** sends stabilizing fibers to its posterior edge.
    - It permits approximately ***6 mm*** of total movement.

- **Lateral Meniscus -** O-shaped.
    - Anchored via the **coronary ligament.**
    - The **popliteus muscle** sends stabilizing fibers to its posterior edge.
    - It permits a much greater range of ***12 mm*** of movement.

- **Meniscopatellar Fibers -** Fibrous bands **connecting the anterior horns of both menisci to the patellar tendon retinaculum.**

- **Meniscofemoral Ligament -** Extends from the **posterior horn of the lateral meniscus to the inner wall of the medial femoral condyle near the PCL.**

Functional Roles

1. Deepens the knee joint to improve lateral and medial stability
2. Absorbs and distributes impact forces by expanding surface contact area
    a. Clinical note: removal increases joint surface stress about threefold
3. Promotes lubrication by spreading synovial fluid across cartilage surfaces
4. Prevents the joint capsule from collapsing or intruding into the joint space

5. Provides partial passive protection against knee hyperextension, with ligaments as the primary defense

---

LIGAMENTOUS SUPPORT OF THE KNEE

- The **collateral ligaments control stability in the medial-lateral (frontal) plane**, while the **cruciate ligaments handle anterior-posterior translation**. Both groups are **extracapsular and become fully taut in extension.**

- **Medial Collateral Ligament (MCL) -** Protects against valgus stress forces in the **frontal plane.**
    - Its posterior section attaches to the medial femoral epicondyle, and its anterior section anchors to the superior aspect of the medial tibia.

- **Lateral Collateral Ligament (LCL) -** Protects against varus stress forces in the **frontal plane.**
    - It attaches proximally to the lateral femoral epicondyle and distally to the lateral aspect of the fibular head, where it blends with the biceps femoris tendon.

- **Anterior Cruciate Ligament (ACL) -** Prevents anterior displacement of the tibia on the femur and protects against hyperextension.
    - Proximally attaches to the anterior tibial fossa (behind the medial meniscus) and distally anchors to the posteromedial aspect of the lateral femoral condyle.

- **Posterior Cruciate Ligament (PCL) -** Prevents posterior displacement of the tibia on the femur and protects against hyperflexion.

    - Proximally attaches to the posterior tibial spine and distally anchors to the anterolateral surface of the medial femoral condyle.

- **Ligament of Humphrey -** Anchors and **stabilizes the lateral meniscus.**
    - It runs from the posterior horn of the lateral meniscus to the distal portion of the PCL attachment on the femur.

- **Ligament of Wrisberg -** Also stabilizes the **lateral meniscus**, running from the posterior horn of the lateral meniscus to the medial femoral condyle.
    - *Anatomical variation:* **It is common to possess either Humphrey or Wrisberg**, but rarely both simultaneously.

TOPIC 3: LEG, ANKLE, AND FOOT

The distal lower extremity is a **biomechanically sophisticated system** that must simultaneously:

- Support **full body weight**
- Provide **shock absorption at initial contact**
- Allow **adaptability on uneven terrain**
- Convert into a **rigid lever for propulsion**
- Maintain **postural equilibrium**

This is achieved through:

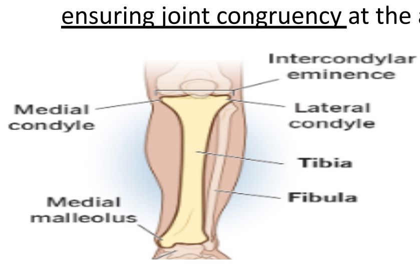
- Structural architecture **(bones + arches)**
- Passive restraints **(ligaments, fascia)**
- Active control **(muscles)**
- Sensorimotor feedback **(proprioception)**

STRUCTURAL ANATOMY

A. TIBIA (Primary Load Transmission)

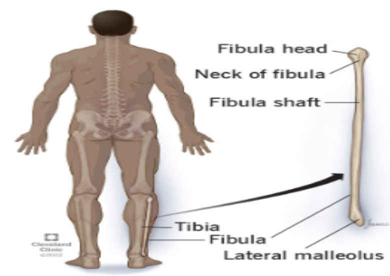
The tibia transmits forces directly **from the femur to the talus.**

- **Tibial Plateau -** Features an **intercondylar eminence, medial condyle, and lateral condyle.**
    - The **medial plateau is larger and concave** to prioritize stability.
    - The **lateral plateau is smaller and convex** to facilitate mobility.
- **Tibial Shaft -** Features a **unique triangular cross-section** that maximizes structural strength against bending forces.
- **Distal Tibia -** Forms the **roof and medial wall of the ankle mortise.**
    - The **medial malleolus** stabilizes the talus medially, handling compressive loads and ensuring joint congruency at the ankle.



B. FIBULA (Stability & Force Distribution)

- Bears only about ***6%--10%*** of total body weight.
- The **fibular head** acts as a lateral stabilizer, while the **shaft, neck, and head serve as muscle attachment sites** and maintain **ankle mortise integrity.**
- The distal end forms the **lateral malleolus**.
- ***Clinical Note:*** Fibular fractures can significantly impair ankle stability despite the bone's minimal weight-bearing role.

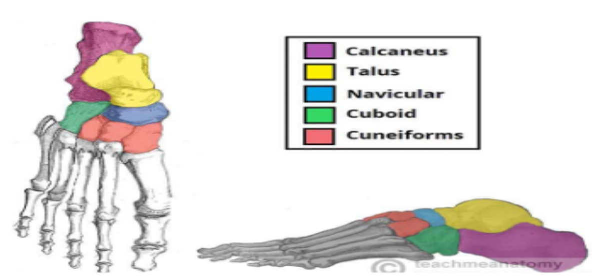


C. TALUS (Central Pivot Bone)

- **No muscular attachments** and is **covered largely by articular cartilage.**
- **Receives force from the tibia** and distributes it down to the calcaneus and forward to the midfoot.
- Acts as the central structural pivot for ankle (talocrural) and subtalar motion.
- ***Clinical Note:*** Highly vulnerable to **avascular necrosis (AVN)** due to its limited, retrograde blood supply.

D. CALCANEUS (Heel Bone)

- The largest tarsal bone; makes the **first contact with the ground during gait.**
    - **Posterior Tuberosity -** Serves as the insertion site for the Achilles tendon (***tendo calcaneus***).
    - **Sustentaculum Tali -** A **medial shelf-like projection that physically supports the talus** and serves as the proximal attachment for the spring ligament.
- **Function - Primary shock absorber of the hindfoot** and acts as a **major lever arm for the plantarflexors.**



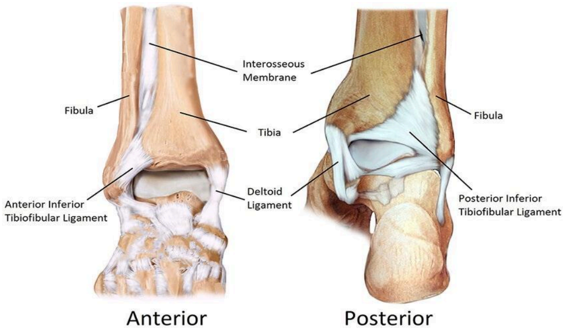
E. MIDFOOT AND FOREFOOT

- **Navicular -** keystone of the medial longitudinal arch.
- **Cuboid -** Stabilizes and supports the **lateral column** of the foot.
- **Cuneiforms (Medial, Intermediate, Lateral) -** Three bones that collectively **form the transverse arch.**
- **Metatarsals (I–V) -** Heads bear weight during the push-off phase of gait. Features a tuberosity on the fifth metatarsal base.
    - **1st Metatarsal -** The stouter, stronger metatarsal, configured to **bear the highest loads during push-off** while working with the sesamoid bones.
    - **2nd Metatarsal -** The longest metatarsal. It acts as the **rigid keystone of forefoot stability** due to its secure, rigid articulation with the cuneiforms.

JOINT COMPLEX AND ARTHROKINEMATICS

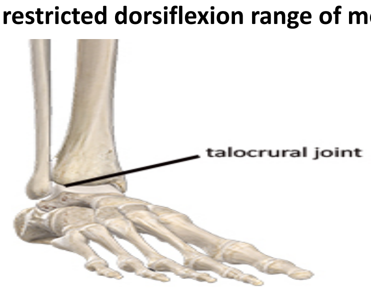
A. DISTAL TIBIOFIBULAR JOINT (SYNDESMOSIS)

- A fibrous joint permitting minimal motion.
- **Function -** Mandates ankle stability while allowing slight widening of the mortise during full dorsiflexion.
    - ***Clinical Note:*** A syndesmotic injury is commonly referred to as a ***"high ankle sprain".***
    - ***Ligaments involved:*** Interosseous membrane, anterior inferior tibiofibular ligament (AITFL), and posterior inferior tibiofibular ligament (PITFL).



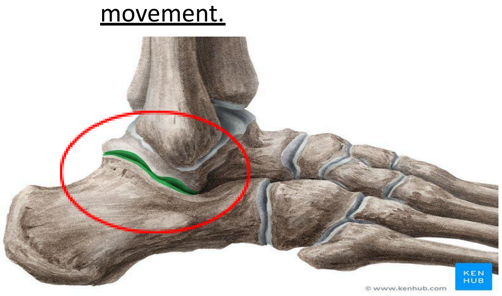
B. TALOCRURAL JOINT (TRUE ANKLE JOINT)

- **Structure -** A mortise **formed by the distal tibia and fibula that grips the convex trochlea of the talus.**
    - The talus is **wider anteriorly** than posteriorly.
- ***Biomechanical Implication:*** The joint is structurally **more stable in dorsiflexion** (as the wider anterior talus wedges into the mortise) and **less stable in plantarflexion**.
- **Arthrokinematics**:
    - **Dorsiflexion:** Anterior roll, posterior glide.
    - **Plantarflexion:** Posterior roll, anterior glide.
    - ***Clinical Application**:* Performing a **passive posterior glide joint mobilization improves restricted dorsiflexion range of motion**.

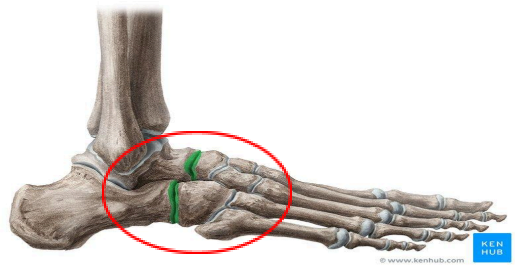


C. SUBTALAR AND MIDTARSAL JOINTS

- **Subtalar Joint -** Functions around an **oblique axis (not purely in the frontal plane).**
    - **Motion -** inversion/eversion with transverse rotation.
    - **Arthrokinematics** - complex triplanar movement.



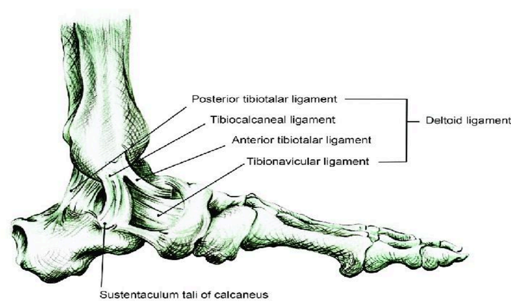
- **MIDTARSAL JOINT (TRANSVERSE TARSAL JOINT)**
    - Composed of the **talonavicular** and **calcaneocuboid articulations**.
    - **Unlocks during pronation** to provide foot flexibility for ground adaptation
    - **Locks during supination** to transform the foot into a rigid lever for propulsion.



LIGAMENTOUS AND FASCIAL FOOT SUPPORT

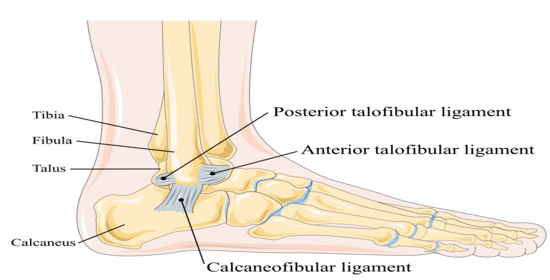
A. MEDIAL SIDE: DELTOID LIGAMENT

- Broad ligament group that **prevents excessive eversion.**
    - Rarely torn completely due to its **inherent strength.**
- **Components:**
    1. Anterior tibiotalar ligament ***(AntaTi)***
    2. Posterior tibiotalar ligament ***(PotaTi)***
    3. Calcaneotibial (tibiocalcaneal) ligament ***(CaTi)***
    4. Tibionavicular ligament ***(TiNa)***



B. LATERAL SIDE LIGAMENTS

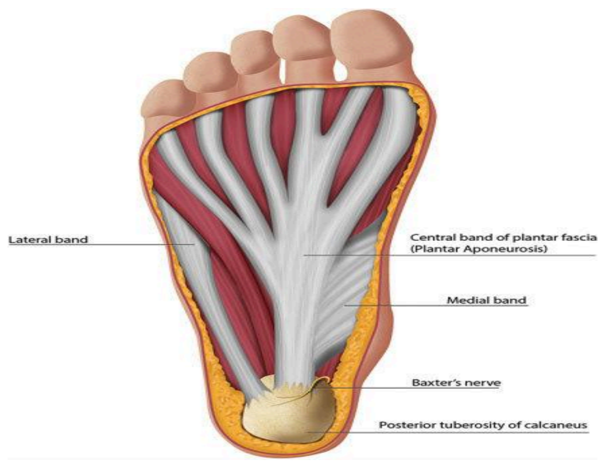
- Significantly **weaker than the deltoid ligament**; guards **against excessive inversion.**
- **Components:**
    1. Anterior talofibular ligament **(ATFL)**
    2. Calcaneofibular ligament (**CFL**)
    3. Posterior talofibular ligament (**PTFL**)
- *Clinical Note:* The **ATFL is the most commonly injured ligament** in the body, typically sprained during a combination of inversion and plantarflexion.



C. PLANTAR FASCIA

- **Plantar Fascia (Plantar Aponeurosis) -** supports the **medial longitudinal arch** and stores/returns elastic energy during gait.
    - **Baxter's nerve** lies in close proximity.

- **Spring Ligament (Plantar Calcaneonavicular Ligament -** Spans from the **sustentaculum tali to the navicular**, supporting the medial arch.
- **Long Plantar Ligament -** Provides primary passive structural support to the lateral longitudinal arch.



MUSCLE FUNCTION AND BIOMECHANICS OF THE LEG/FOOT

- **DORSIFLEXORS:**
  - **Tibialis Anterior** - primary dorsiflexor and a strong inverter.
    - It works via **eccentric control during heel strike** to decelerate the foot and prevent a sudden ***"foot slap".***
- **PLANTARFLEXORS:**
  - **Gastrocnemius** - **fast-twitch fibers** optimized for rapid power generation
  - **Soleus** - **slow-twitch fibers** optimized for sustained postural control.
  - They generate the vital push-off force during terminal stance, contributing up to ***80%--85%*** of forward propulsion.
- **INVERTORS:**
  - **Tibialis anterior and tibialis posterior.**
  - They act **to maintain the medial longitudinal arch** and control **pronation**
- **EVERTORS:**
  - **Fibularis (peroneus) longus, fibularis brevis**
  - They provide **essential lateral stability** and **prevent excessive or harmful inversion.**

METATARSALS

GENERAL FEATURES

Total: **5 metatarsals (MT I–V)**

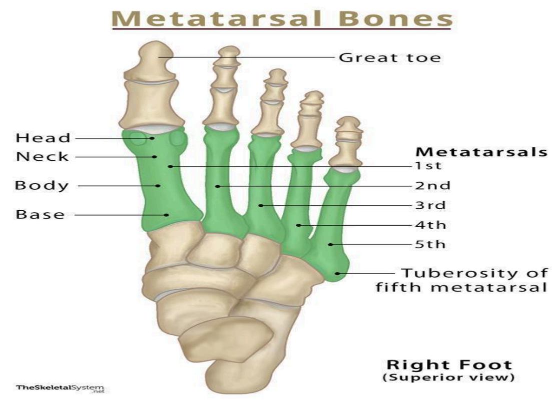
- Structure:
  - **Base (proximal)**
  - **Shaft**
  - **Head (distal)**

IMPORTANT TRIVIA

- **Longest -** 2nd metatarsal
- **Stoutest/Strongest -** 1st metatarsal
  - Reason: **bears the highest load during push-off**

BIOMECHANICAL ROLE

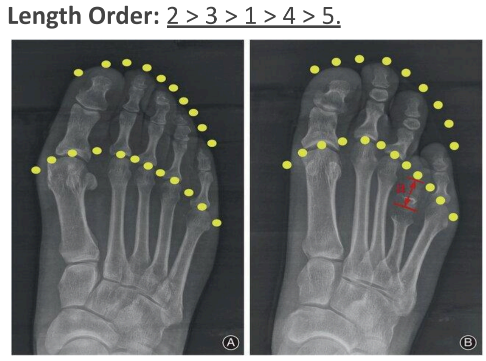
- **1st MTT -** Primary weight-bearing during terminal stance
  - Works with sesamoids for push-off
- **2nd MTT -** Acts as a **keystone of forefoot stability**
  - **Most rigid** due to cuneiform articulation



FOOT CONFIGURATION TYPES

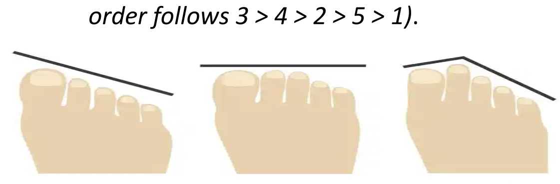
A. Metatarsal Head Projection

a. **Length Order:** 2 > 3 > 1 > 4 > 5.



B. Toe Length Patterns

- **Morton's Foot (Grecian / Atavistic ~23%):**
  - 2 > 1 > 3 > 4 > 5.
  - *Clinical note:* Results in **localized stress and pain over the 2nd metatarsal head.**
- **Egyptian Foot (Most Common ~69%):**
  - 1 > 2 > 3 > 4 > 5.
- **Roman Foot:**
  - 1 = 2 = 3 > 4 > 5.
- *(Hand Comparison Metric: Finger length order follows 3 > 4 > 2 > 5 > 1).*



ARCHES OF THE FOOT

1. **MEDIAL LONGITUDINAL ARCH**

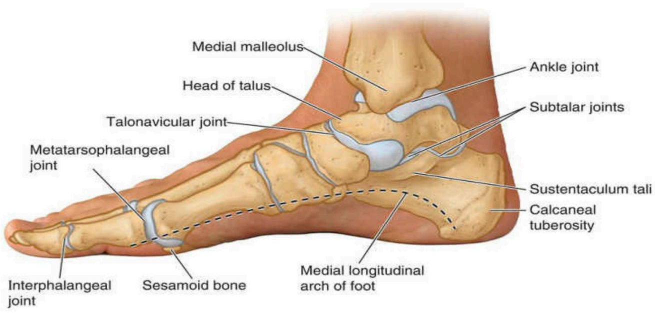
- **BONES**
  - Calcaneus
  - Talus (**keystone: head of talus**)
  - Navicular (**most important clinically**)
  - Cuneiforms (1–3)
  - Metatarsals (1–3)
- **LIGAMENT SUPPORT**

- **Spring ligament (Plantar calcaneonavicular ligament)**
  - Proximal attachment: **sustentaculum tali**

- <u>**DYNAMIC SUPPORT**</u>
  - **Tibialis posterior tendon (most important)**

- <u>**SPECIAL TEST**</u>
  - **Navicular Drop Test (Feiss Line)**
    - Line: **Medial malleolus → 1st MTP**
    - Normal:
    - Navicular lies on or slightly above line
    - Positive:
    - Navicular drops below line → <u>**pes planus**</u>

- <u>**CONDITIONS**</u>
- <u>**Pes Planus (Flat Foot)**</u>
  - Characteristics:
    - Pronated foot
    - **Collapse of medial arch**
    - Associated with:
    - **Tibialis posterior weakness**
    - **Ligament laxity**
  - Also called as **Pes valgus**
- <u>**Pes Cavus (High Arch)**</u>
  - Characteristics:
    - Supinated, rigid foot
    - **Poor shock absorption**
    - Common in:
    - **Charcot-Marie-Tooth disease**
    - Classic sign:
    - ***"Inverted champagne bottle"*** (leg muscle wasting)



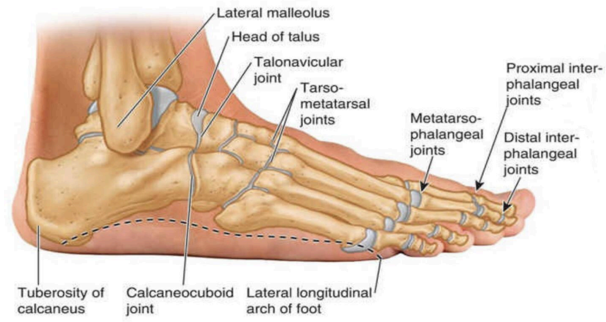
2. <u>**LATERAL LONGITUDINAL ARCH**</u>

- <u>**BONES**</u>
  - Calcaneus
  - Cuboid (**keystone**)
  - **4th and 5th metatarsals**

- <u>**LIGAMENT**</u>
  - Long plantar ligament

- <u>**DYNAMIC SUPPORT**</u>
  - Fibularis (Peroneus) longus tendon

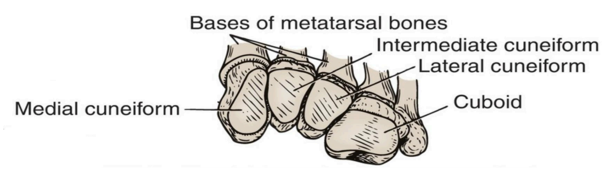
- <u>**CLINICAL ASSOCIATION**</u>
  - <u>**Pes varus**</u>
    - Supinated, inverted foot



3. <u>**TRANSVERSE ARCH**</u>

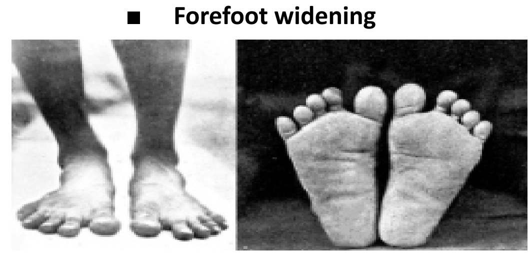
- <u>**BONES**</u>
  - Cuneiforms (3)
  - Cuboid
  - Bases of metatarsals
  - **Keystone: 2nd cuneiform**

- <u>**SUPPORT**</u>
  - **Intrinsic foot muscles <u>(primary)</u>**
  - **Fibularis longus + tibialis posterior <u>(secondary)</u>**

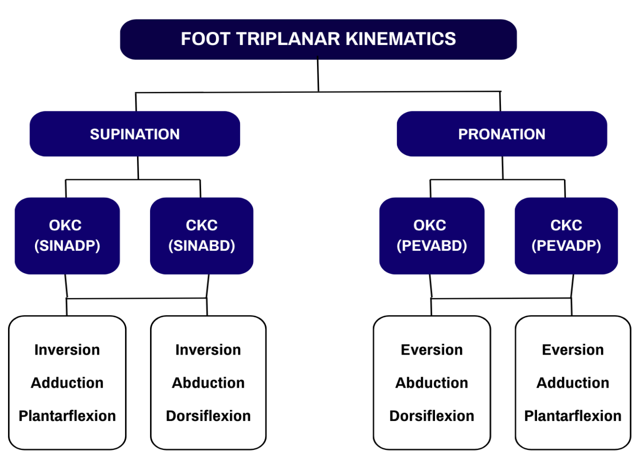


- **CONDITION**

  <u>**Splay Foot**</u>
  - Cause:
    - **Weak intrinsic muscles**
  - Result:
    - **Collapse of transverse arch**
    - **Forefoot widening**



PRONATION AND SUPINATION



<u>**SINUS TARSI**</u>

- Contains:
  - **Interosseous talocalcaneal ligament**
- Function - **Proprioception; Stability**
- Clinical:
  - ***Sinus Tarsi Syndrome*** → instability, pain

MUSCLES OF THE FOOT

DORSAL COMPARTMENT

- **Extensor Digitorum Brevis (EDB)**
  - **Only intrinsic muscle on dorsum**
  - No counterpart in hand

PLANTAR COMPARTMENT (4 LAYERS)

1. **Layer 1 (Medial)**
   - Abductor hallucis
   - Flexor digitorum brevis
   - Abductor digiti minimi

2. **Layer 2 (Central/Master Knot of Henry)**
   - Lumbricals
   - Quadratus plantae

3. **Layer 3 (Lateral)**
   - Flexor hallucis brevis
   - Adductor hallucis
   - Flexor digiti minimi brevis

4. **Layer 4 (Interosseus)**
   - Interossei (7 total)
   - 4 dorsal (abduct)
   - 3 plantar (adduct)

NERVE SUPPLY

A. **TIBIAL NERVE**

1. **Medial Plantar Nerve ("LAFF" muscles)**
   - **L**umbrical (1st)
   - **A**bductor hallucis
   - **F**lexor hallucis brevis
   - **F**lexor digitorum brevis

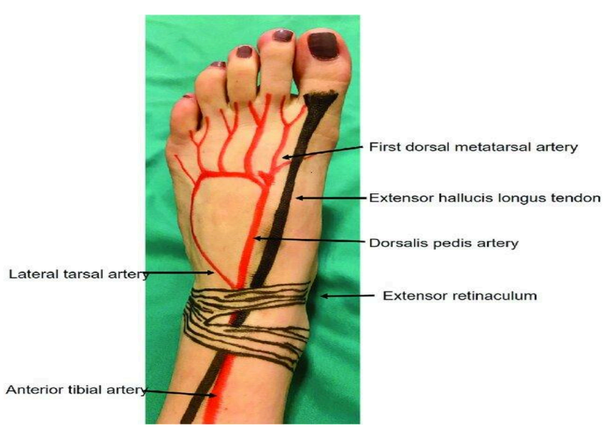
2. **Lateral Plantar Nerve**
   - All other intrinsic muscles

BLOOD SUPPLY

- **Arterial Flow:**
- Femoral artery
  → Popliteal artery
  → Anterior tibial artery
  → **Dorsalis pedis artery**

DORSALIS PEDIS PALPATION

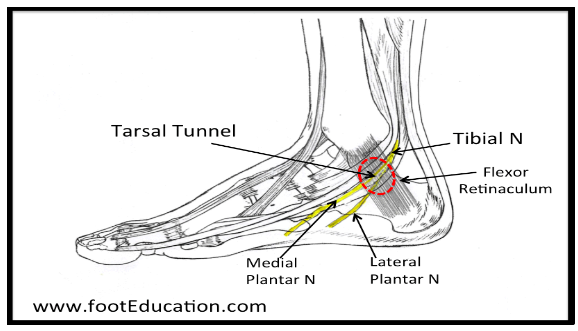
- Lateral to tibialis anterior
- Lateral to extensor hallucis longus
- Between 1st and 2nd metatarsals
- Mnemonic:
  - Between **EDL and EHL**



RETINACULA AND TUNNELS

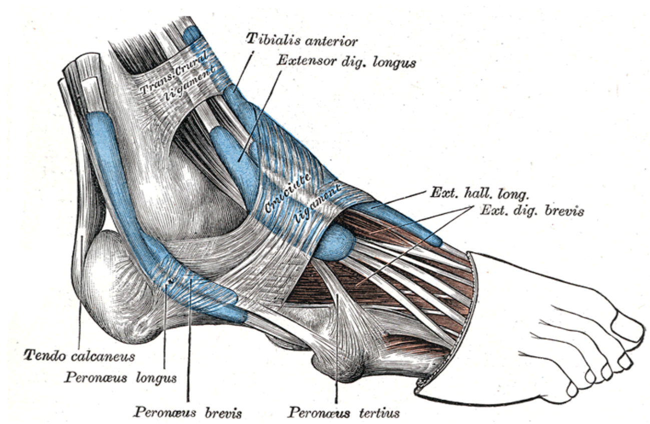
A. **TARSAL TUNNEL/ FLEXOR RETINACULUM (MEDIAL)**

- Covered by:
  - Flexor retinaculum
- Contents (Mnemonic: **Tom, Dick, And Very Nervous Harry**)
  - **Tibialis** posterior
  - Flexor **digitorum** longus
  - Posterior tibial **artery**
  - Posterior tibial **vein**
  - Tibial **nerve**
  - Flexor **hallucis** longus
- **Clinical:**
  - Tarsal Tunnel Syndrome
  - Symptoms:
    - Sole numbness
    - Tingling



B. **EXTENSOR RETINACULUM**

- Mnemonic:
  - **Tom Has A Very Nice Dog, Elmo & Fred.**
    - **Tibialis** anterior
    - Extensor **hallucis** longus
    - **Anterior** tibial artery
    - **Vein**
    - Deep fibular **nerve**
    - Extensor **digitorum** longus
    - Fibularis **tertius**



TOPIC 4: GAIT ANALYSIS

GAIT - manner or pattern of walking of an individual.

- Process by which the **body is propelled forward through the coordinated actions of the lower extremities.**
  - Uncoordinated movements can lead to abnormal patterns such as ***limping, staggering, or excessive swaying.***
- Considered a **highly coordinated motor activity.**
- **Forward propulsion** - body is being pushed or moved forward.

- **Nervous system** - continuously monitors the body position and movement through **sensory feedback.**

Semi-Rotatory Movements - natural rotational motions of head, trunk, and arms while walking.

- **Reciprocal Arm Swing** - when the arm swing alternates with our legs.
    - Helps maintain balance and counteracts, of course, the rotational forces produced by our lower limbs.
    - Trunk also rotates slightly to assist with smooth weight transfer from one side of the body to another.
    - Head remains relatively stable to maintain visual focus and equilibrium while walking.

Springiness Of The Feet

- **Arches** - natural **shock absorbers.**
    - Compress slightly upon contact with the ground and then release stored energy to assist in forward propulsion.

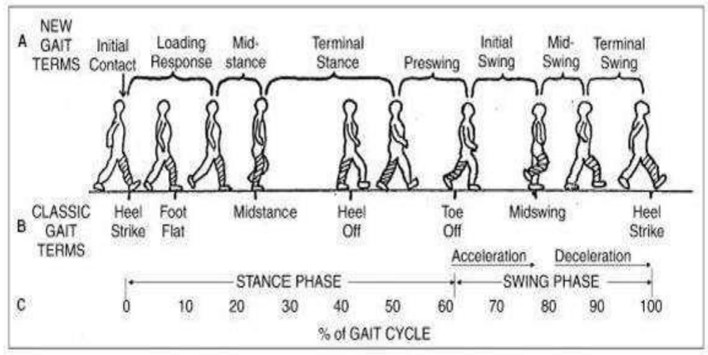
GAIT CYCLE

- Definite cyclic order, meaning the movement pattern repeats continuously.
- **The basic unit of measurement in gait analysis.**

2 Major Phases:

1. **STANCE PHASE** - occurs when the foot is **in contact with the ground supporting our body weight**
    - Occupies about ***60%*** of the entire gait cycle.
    - **Weight acceptance** - first part of stance phase.
        - When the foot initially contacts the ground and starts accepting the body weight.
        - Both feet are briefly on the floor, which is known as the ***double support*** (provides additional stability); occurs twice in one gait cycle.
    - **Single Limb Support Phase** - only one lower extremity supports the entire body weight.
        - **Hip abductors, quads, and plantarflexors** work actively to prevent collapse and maintain smooth progression.

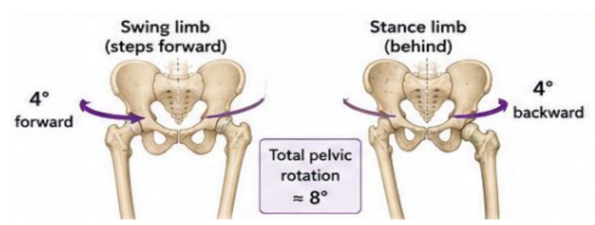
2. **SWING PHASE** - occurs when the **foot is lifted off the ground and moves forward the next step.**
    - Occupies ***40%*** of the whole gait cycle.
    - Main purpose - **limb advancement**.
    - **Hip flexors** help move the thigh forward; the **knee** will flex to allow foot clearance; the **ankle dorsiflexors** prevent the toes from dragging on the floor.



LOWER EX MECHANISMS

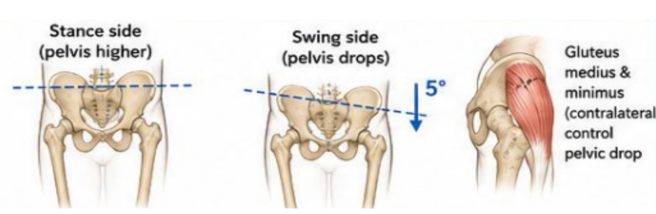
1. **Pelvic Rotation (Transverse Plane)**

- Pelvis rotates approximately ***4°*** backward on the stance limb; total of ***8°***
- **Purpose:**
    - Increase the effective leg length
    - Reduces the amount of hip flexion and extension needed
    - **Smooths out vertical COG displacement**



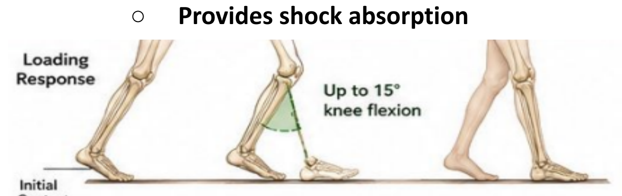
2. **Pelvic Tilt (Frontal Plane)**

- Pelvis drops slightly about ***5°*** on the side of the swinging leg
- **Purpose:**
    - **Reduces the rise of the COG during midstance**
    - Controlled by eccentric contraction of the contralateral gluteus medius and minimus
    - **Clinical Note:** If weak, leads to **Trendenlenburg Gait**



3. **Knee Flexion during Stance Phase**

- Knee flexes up to ***15°*** during loading response
- **Purpose**:
    - Lowers the body slightly at initial contact
    - **Decreases the peak of the vertical COG displacement**
    - **Provides shock absorption**



4. **Foot and Ankle Mechanism**

- Includes **heel strike** (controlled plantarflexion) and **toe-off** (push off)
- Purpose:
    - Smooths the **vertical trajectory of the COG**
    - Facilitates forward propulsion
    - Absorbs shock and helps lengthen the limb functionally

5. **Knee-Ankle Interaction**

- During terminal stance, the ankle dorsiflexors contact while the knee extends
- **Purpose:**
    - Lengthens the leg for efficient push-off
    - Assists in smooth lowering of the body
    - Helps conserve momentum and energy

6. **Lateral Displacement of the Pelvis**

- Pelvis shifts laterally about ***2-5cm*** toward the stance limb
- **Purpose:**
    - Maintains balance over the stance leg
    - Minimizes excessive vertical movement
    - **Controlled by Hip Abductors**

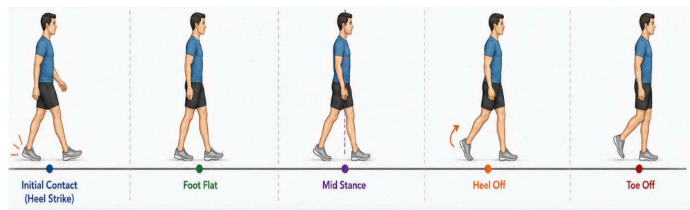
A. Pelvic Tilt - Pelvis tips downward on the swing side, weakness can lead to **Trendelenburg Gait.**

B. Pelvic Displacement - Pelvis and upper body shift sideways toward the stance leg.

- Excessive shift may be seen in compensated **Trendelenburg Gait** or **Waddling Gait**

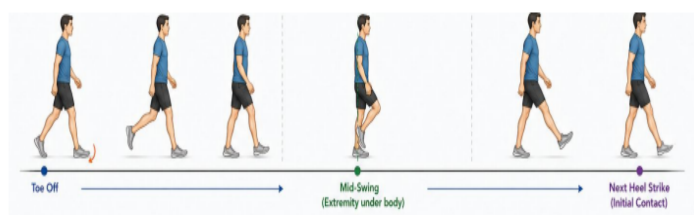
STANCE PHASE

HEEL STRIKE PHASE	FOOT FLAT	MID-STANCE	HEEL OFF	TOE OFF
Begins with **initial contact and ends with foot flat.** Beginning of stance phases **when the heel contacts the ground.**	Occurs immediately following heel strike. Point where the **foot fully touches the ground.**	Body passes directly over the supporting extremity.	Point following midstance; **heel of the reference extremity leaves the ground.**	Point following heel off when the **toe of the reference extremity is in contact with the ground.**



SWING PHASE

ACCELERATION PHASE	MID-SWING	DECELERATION
Begins **once the toe leaves the ground and continues until mid-swing**; swinging extremity is directly under the body.	When the **body passes directly beneath the body/from the end of acceleration to the beginning of deceleration.**	After mid-swing, **when the limb is decelerating in preparation for heel strike.**



RANCHO LOS AMIGOS (RLA) PHASES OF GAIT

Stance Phase (60% of Gait Cycle)

1. Initial Contact (0% — Initial Double Limb Stance)

- **Definition -** The initial contact of the foot of the
    - Normally, the **heel is the first part to touch the ground** (heel strike).
- **Clinical Note -** In abnormal gait, the whole foot or the toes may strike the ground first instead of the heel.

2. Loading Response (0%–11% — Double Limb Stance)

- **Definition -** Begins at initial contact and ends when the contralateral (opposite) extremity lifts off the ground at the end of the double-support phase.
- **Purpose -** Shock absorption, weight acceptance, and maintaining forward progression.

3. Mid-Stance (11%–30% — Single Limb Stance)

- **Definition -** Begins when the contralateral extremity lifts off the ground and ends when the body is directly over the supporting limb.
- **Purpose -** Single limb support, body progression over a stable foot, and maintaining balance and stability.

4. Terminal Stance (30%–50% — Single Limb Stance)

- **Definition -** Begins when the body is directly over the supporting limb and ends just before initial contact of the contralateral extremity.

- **Purpose -** Forward propulsion, heel rise occurrence, and preparation for weight transfer.

5. Pre-Swing (50%–60% — Terminal Double Limb Stance)
- **Definition -** The last 10% of the stance phase. It begins with the initial contact of the contralateral foot and ends with toe-off.
- **Purpose -** Rapid unloading of the limb, preparation for the swing phase, and push-off generation.

Swing Phase (40% of Gait Cycle)
6. Initial Swing (60%–73%)
- **Definition -** Begins when the toe leaves the ground and continues until maximum knee flexion occurs.
- **Purpose -** Foot clearance, forward limb acceleration, and advancement of the extremity.

7. Mid-Swing (73%–87%)
- **Definition -** Encompasses the period from maximum knee flexion until the tibia is in a vertical position.
- **Purpose:** Continued limb advancement, adequate foot clearance, and controlled forward momentum.

8. Terminal Swing (87%–100%)
- **Definition -** Includes the period from the point where the tibia is vertical to a point just before initial contact.
- **Purpose -** Limb deceleration, preparation for heel strike, and proper positioning for weight acceptance.

Clinical Correlations & Problems
- **Initial Contact Abnormalities -** ***Forefoot strike*** may indicate plantarflexor spasticity or equinus gait.
  - ***Flat foot contact*** - weak dorsiflexors.
- **Loading Response Problems -** Weak quadriceps lead to *knee buckling*.
  - Pain during weight acceptance leads to an ***antalgic gait*.**
- **Mid-Stance Problems -** Hip abductor weakness leads to ***Trendelenburg gait***, causing poor balance and stability.
- **Terminal Stance / Pre-Swing Problems -** Weak plantarflexors lead to ***poor push-off*** and decreased propulsion.
- **Swing Phase Problems -** *Foot drop* leads to a ***steppage gait*.**
  - Can result in circumduction gait, hip hiking, poor foot clearance, and an increased tripping risk.

TOE-OUT ANGLE (FOOT PROGRESSION ANGLE)

- Angle formed **between the line of progression during walking** and the **long axis of the foot.**
- Normal: foot points **<u>slightly outward</u>** during gait.

Normal values
- ***5-7 degrees*** external rotation
- May normally range from ***0-15 degrees***
- Angle is **measured during stance phase of gait**

Purpose/Function
- **Improve balance and stability**
  - **Slight toe-out** - widens base of support during walking
- Assists forward propulsion, weight transfer, and compensates for limb alignment

<u>Type of Foot Progression</u>

NORMAL TOE-OUT	EXCESSIVE TOE-OUT	TOE-IN GAIT
• Slightly outward • ***5-7 degrees ER***	• ***"Duck-like"*** walking pattern • External tibial torsion, hip ER contracture • Flat feet, weak hip stabilizers • ***Greater than 15 degrees***	• Inward feet, causing tripping tendency • Knees and feet appear IR • Femoral anteversion • Internal tibial torsion • Metatarsus adductus • Neurologic conditions (CP, Parkinson's Disease, Muscular Dystrophy) • ***<0 degrees***

SPATIAL PARAMETERS OF GAIT

- Distance measurements involved in walking

1. **STEP LENGTH** - distance from **one foot to the opposite foot**
   - ***35-41 cm***
   - Equal on both sides during gait
   - Reflects symmetry and efficiency of gait

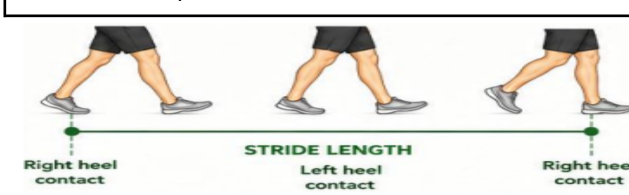
<u>Clinical significance:</u>

Shortened step length	Asymmetrical step
- Parkinson's disease - Painful lower ex conditions - Weakness	- Antalgic gait - Neurologic disorders - Muscle weakness

2. **STRIDE LENGTH -** distance of **same foot to same foot (one right step + one left step)**
   - ***70-82 cm***
   - Walking speed and mobility

<u>Clinical significance:</u>

Reduced stride length may occur in: - Shuffling gait - Parkinsonian gait - Elderly individuals



3. **STEP WIDTH (WALKING BASE)** - **side-to-side distance** between feet
   - ***5-10 cm***
   - Balance and postural stability

Clinical significance:

Increased Step Width	Decreased Step Width
- Ataxic gait - Cerebellar disorders - Poor balance	- Tightrope-like gait - Spastic gait patterns

__WIDTH OF BASE SUPPORT__ - distance between midpoints of two feet ***(5-10 cm)***

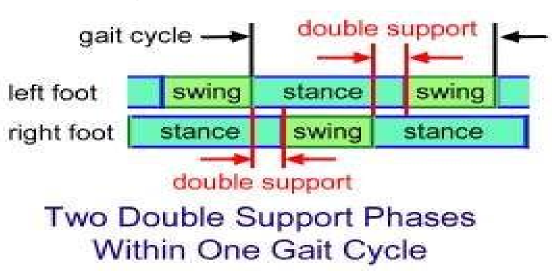
__CADENCE__ - number of steps taken per minute (varies in step length, speed of walking, sex, body built, surface, etc.)

GAIT TIMING PARAMETERS

1. **STANCE TIME** - amount of time elapses during the **__stance phase of one extremity in gait cycle.__**

2. **SINGLE-SUPPORT TIME** - amount of time elapses during the period **__when only one extremity is on the supporting surface in a gait cycle.__**

3. **DOUBLE-SUPPORT TIME** - when **__both legs support the body weight__**
   - Inversely **proportional to the cadence of gait**



MUSCLE ACTIVITY IN GAIT

- **__ANKLE AND FOOT__**

PHASE	Joint Position	Joint Moment	Muscular Activit
Initial Contact	Neutral	Going Dorsiflexion	TA, ED, EHL
Loading Response	5 degree plantarflexion	Maximum Dorsiflexion	TA, ED, EHL

PHASE	Joint Position	Joint Moment	Muscular Activity
Initial Contact	Neutral	Going Dorsiflexion	TA, ED, EHL
Loading Response	5 degree plantarflexion	Maximum Dorsiflexion	TA, ED, EHL

PHASE	Joint Position	Joint Moment	Muscular Activity
Mid Stance	5 degree plantarflexion	Going Plantarflexion	Gastroc-solues group
Terminal Stance	10 degree dorsiflexion	Maximum Plantarflexion	Gastroc –solues group

PHASE	Joint Position	Joint Moment	Muscular Activity
Pre – Swing	15 degree plantarflexion		Decreasing contraction of Gastroc-Soleus Group

PHASE	Joint Position	Joint Moment	Muscular Activity
Initial Swing	5 degree plantarflexion	Low dorsiflexor moment	TA, ED, EHL
Mid Swing	Neutral	Low dorsiflexor moment	TA, ED, EHL
Terminal Swing	Neurtral	Low dorsiflexor moment	TA, ED, EHL

- KNEE

PHASE	Joint Position	Joint Moment	Muscular Activity
Initial Contact	Fully extended	Brief Flexor Moment	Hamstrings RESISTS hyperextension
Loading Response	20 degrees flexion	Extensor Moment	Vastii activity allows knee flexion for shock absorption however prevents collapse

PHASE	Joint Position	Joint Moment	Muscular Activity
Mid Stance	Appears Fully Extended	Extensor to Flexor Moment	Vastii slowly stops contracting
Terminal Stance	Appears Fully Extended	Flexor Moment	Vastii slowly stops contracting

PHASE	Joint Position	Joint Moment	Muscular Activity
Pre Swing	40 degrees flexion	Extensor Moment	Rectus Femoris Modulates Rate of Knee Flexion
Initial Swing	60 degrees flexion	Extensor Moment	Biceps Femoris, Gracilis, Sartorius contribute to knee flexion

PHASE	Joint Position	Joint Moment	Muscular Activity
Mid Swing	25 degree flexion	Flexor Moment	Hamstrings modulate rate of knee extension
Terminal Swing	Appears fully extended	Flexor Moment	Hamstrings continue activity and vastii prepares for initial double limb stance

- **__HIP__**

PHASE	Joint Position	Joint Moment	Muscular Activity
Initial Contact	20 degrees flexion	Extensor Moment	Gluteus Maximus, Medius and Minimus stabilizes trunk over femur
Loading Response	20 degrees flexion	Extensor Moment	Gluteus Maximus, Medius and Minimus stabilizes trunk over femur

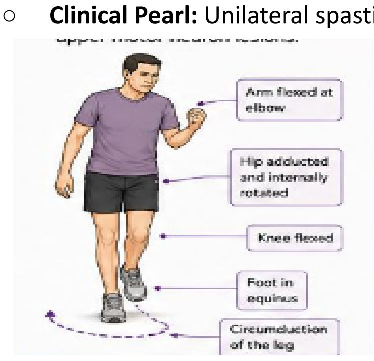
PHASE	Joint Position	Joint Moment	Muscular Activity
Mid Swing	25 degrees flexion	Extensor Moment	Increasing hamstring activity to restrain thigh advancement
Terminal Swing	20 degrees flexion	Extensor Moment	Hamstrings control thigh posture while hip extensors and abductors increase in activity.

COMMON GAIT ABNORMALITIES

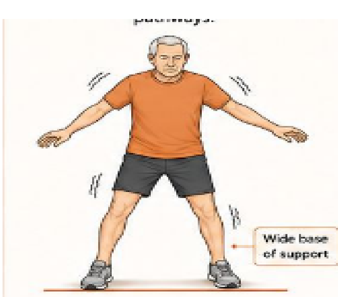
1. **ANTALGIC GAIT**
    - **Cause:** Pain in weight-bearing limb (arthritis, fracture, sprain).
    - **Feature:** Shortened stance on painful side, reduced weight bearing.
    - **Clinical Pearl:** Pain-avoidance gait.

2. **TRENDELENBURG GAIT**
    - **Cause:** Hip abductor weakness (gluteus medius/minimus, L5 lesion).
    - **Feature:** Pelvis drops on swing side, trunk leans toward stance limb.
    - **Clinical Pearl:** Hip abductor insufficiency.

3. **HEMIPLEGIC GAIT**
    - **Cause:** Stroke, cerebral palsy, brain/spinal injury.
    - **Feature:** Circumduction of affected leg, flexed arm, equinus foot.
    - **Clinical Pearl:** Unilateral spastic weakness.



4. **ATAXIC GAIT**
    - **Cause:** Cerebellar/vestibular dysfunction, neuropathy, MS, intoxication.
    - **Feature:** Wide base, staggering, irregular stride, poor coordination.
    - **Clinical Pearl:** Loss of balance/coordination.



5. **SHUFFLING GAIT**
    - **Cause:** Parkinson's disease, NPH, dementia.
    - **Feature:** Short sliding steps, festination, reduced arm swing, en bloc turning.
    - **Clinical Pearl:** Difficulty initiating movement.